

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |       |  | A. Xylem contents under lower pressure (than air) (1)<br>B. contents pulled (upwards through the xylem) due to {transpiration stream / cohesion-tension / adhesion/ evaporation of water vapour from the leaves} (1)<br>C. Phloem contents under higher pressure (than air) (1)<br>D. contents {pushed/forced} through phloem by {mass flow/ active transport/ hydrostatic pressure} (1) |                 | 4   | 0   | 4     |       | 4    |
|          | (c) | (i)   |  | Amino acid/named amino acid / (named) hormone (1)                                                                                                                                                                                                                                                                                                                                        | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | <b>Any four (x1) from:</b><br>A. There are (high levels of) radioactivity in young leaves (1)<br>B. Young leaves are sinks (1)<br>C. so {using/importing} sucrose/ sucrose transported to young leaves (1)<br>D. (as an energy source) for {growth/cell division} (1)<br>E. Lack of radioactivity in older leaves as {they are sources/ produce their own sucrose} (1)                   |                 | 2   | 2   | 4     |       | 4    |
|          |     | (iii) |  | Bidirectional flow/ both up and down (the stem)/ both directions                                                                                                                                                                                                                                                                                                                         |                 | 1   |     | 1     |       | 1    |